

Project Report: Diamond Price Prediction Using Regression

Problem Statement

Pricing a diamond accurately depends on a combination of physical and quality attributes — carat weight, cut, color, clarity, and dimensions — that interact in complex, often non-linear ways. Manual or rule-based pricing is difficult to scale and prone to inconsistency. This project aims to build a reliable regression-based system to predict diamond prices from their measurable attributes, while also investigating a common data-quality issue (multicollinearity) that can silently distort model interpretation if left unaddressed.

Project Overview

We developed and compared multiple regression models that analyze diamond attributes and predict price in US dollars. Beyond standard model comparison, this project includes a deeper investigation into whether regularization techniques (Ridge, Lasso, Elastic Net) genuinely resolve multicollinearity between correlated features, or merely appear to while leaving coefficient interpretation unreliable.

Dataset Summary

The dataset (diamonds.csv) includes the following key features:

- Carat (weight of the diamond)
- Cut (quality grade: Fair to Ideal)
- Color (grade: J worst to D best)
- Clarity (grade: I1 worst to IF best)
- Depth (total depth percentage)
- Table (width of top facet, as % of diameter)
- x, y, z (length, width, depth in mm)

The target variable is:

- Price (USD)

The dataset was largely clean but contained a small number of data-entry errors — 17 rows with zero-value dimensions and 6 rows with internally inconsistent x/y/z measurements (e.g., one dimension several times larger than the other two for the same diamond). These were identified using a data-driven ratio check and removed. Categorical features (cut, color, clarity) were ordinally encoded to preserve their natural quality ranking. Statistically rare but valid values (large, expensive diamonds) were deliberately retained, since removing them would discard genuine market information.

Exploratory Data Analysis (EDA Highlights)

- Price is heavily right-skewed: most diamonds fall between \$300–\$2,000, with a long tail up to \$18,823.
- Carat shows the strongest relationship with price (correlation: 0.92), followed by the x, y, z dimensions (0.87–0.89).
- Depth and table showed negligible linear relationship with price (correlation near 0).
- Carat, x, y, and z are severely inter-correlated (0.98–0.99), indicating significant redundancy among these features.
- Counter-intuitively, higher color/clarity grades showed a weak negative correlation with price — larger rough stones are often cut into lower-grade finished diamonds, so this reflects a genuine market pattern rather than an error.

Model Development

We trained multiple models including:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Elastic Net Regression
- Random Forest Regression

Random Forest yielded the best results with:

- R^2 Score: ~0.982
- RMSE: ~\$527

This made it the strongest choice for prediction accuracy. However, Linear Regression, Ridge, and Elastic Net all performed comparably well ($R^2 \approx 0.95$) once price was log-transformed for training, and Random Forest's improvement over these was found to be modest but consistent.

We used a structured pipeline to:

- Clean and validate the raw data
- Encode ordinal categorical features
- Scale numeric features for the linear-family models

- Log-transform the target variable (Linear/Ridge/Lasso/Elastic Net only) to counter price's skew
- Tune hyperparameters via cross-validation for every regularized model and Random Forest

Evaluation Metric

To evaluate model accuracy and robustness, we used:

- R^2 Score (variance explained)
- RMSE (Root Mean Squared Error, in dollars)
- MAE (Mean Absolute Error, in dollars)

RMSE and MAE were reported in real price units (not log-scale) by reversing the log-transform on predictions before evaluation, ensuring every model was compared fairly on the same scale.

Multicollinearity Investigation

A central part of this project examined whether Ridge, Lasso, and Elastic Net actually resolve the severe multicollinearity found between carat, x, y, and z (VIF values of 25 to 885). Using carat's regression coefficient — which should always be positive — as a concrete test:

- Carat's coefficient was strongly positive (+0.93) when modeled alone, but flipped negative (-0.53) once modeled alongside x, y, z.
- Ridge, Lasso, and Elastic Net, when each properly tuned via cross-validation for best predictive accuracy, all failed to correct this sign distortion.
- Only heavy, non-accuracy-optimal regularization (Lasso at a high alpha) resolved the sign issue, at a steep cost to accuracy (R^2 fell from 0.95 to 0.78).
- Removing the redundant x, y, z features directly restored a correct, positive coefficient for carat (+1.03), without needing any regularization at all.

This shows that regularization tuned for prediction accuracy does not automatically fix severe multicollinearity's effect on coefficient interpretability — a structural fix was required instead.

Challenges

- Distinguishing genuine data errors from rare-but-valid extreme values required a data-driven diagnostic (a ratio-based check) rather than an arbitrary cutoff.

- Severe multicollinearity between carat and the dimensional features made coefficient interpretation unreliable and required a dedicated investigation to resolve properly.
- Balancing predictive accuracy against coefficient interpretability surfaced a genuine trade-off rather than a straightforward fix.

Impact

This project can:

- Serve as a pricing reference tool for diamond buyers, sellers, or appraisers
- Help identify which diamond attributes most influence price, once multicollinearity is properly accounted for
- Serve as a learning example of how regularization does and does not solve real-world multicollinearity problems

Tech Stack

- Python
- Scikit-learn
- Pandas, NumPy
- Matplotlib, Seaborn
- Statsmodels (VIF calculation)

Future Work

- Deploy the best-performing model (Random Forest) via a simple web interface for interactive price prediction
- Add SHAP-based explainability for the Random Forest model
- Explore gradient boosting models (XGBoost, LightGBM) for potential further accuracy gains
- Extend the multicollinearity investigation to other correlated feature groups if the dataset is expanded